

RAVITEJA MADDELA

MSc Computer Science & Engineering Student

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teja-mr-1.github.io/portfolio

PROFESSIONAL SUMMARY

MSc Computer Science student at the Technical University of Denmark (DTU) with prior software development experience across healthcare, insurance, and financial domains (GE Healthcare, Swiss Re, American Express). Currently focusing on specializations in Artificial Intelligence, Machine Learning, and Data Science during MSc studies at DTU.

EDUCATION

Technical University of Denmark (DTU) Sep 2025 – Present
Master of Science in Computer Science & Engineering (Copenhagen, Denmark)
Key Focus: Artificial Intelligence, Machine Learning, and Data Science. Student ID: s252744.

RVR & JC College of Engineering (Acharya Nagarjuna University)
Bachelor of Technology in Information Technology (India)

TECHNICAL SKILLS

AI & Machine Learning:	PyTorch, QLoRA, Transformers, RAG, Mixture-of-Experts (MoE), BioClinicalBERT, FAISS, LLMs, MCP, WebMCP
Data Engineering & Cloud:	Python, SQL, PostgreSQL, DuckDB, Apache Parquet, Pydantic, Power BI, AWS, Docker, Git, GitHub Actions
Software & Web:	Java, JavaScript, FastAPI, Spring Boot, REST APIs, Leaflet.js, Mapbox, React, Tailwind CSS

FEATURED ENGINEERING & RESEARCH PROJECTS

Sentinel: Data Quality Pipeline & AI Assistant Python Pydantic DuckDB Power BI Personal Data Engineering Project

- Engineered a modular data ingestion pipeline featuring 4-tier validation (Pydantic schema checks, SHA-256 deduplication, Z-score outlier detection).
- Built an interactive Streamlit Quarantine Remediation Console to edit and re-inject failed records live, exporting clean datasets to Apache Parquet and Power BI.
- Integrated an AI Telemetry Assistant using OpenRouter API to converse with live pipeline telemetry metrics.

Sparse MoE-QLoRA for Danish Legal QA PyTorch MoE QLoRA Personal Research Project

- Built a 4-bit Quantized Low-Rank Adaptation (QLoRA) pipeline to fine-tune a 7B model for Danish legal question answering.
- Implemented centroid-based Sparse Mixture-of-Experts (MoE) routing using E5 embeddings paired with RAG document retrieval.
- Tested memory pre-caching and in-place tensor copy to enable fast expert adapter switching on a single T4 GPU.

Weather-Aware Route Planner & Travel Advisor JavaScript Leaflet.js Mapbox Personal Project

- Built an interactive route planning web application combining Leaflet.js and Mapbox GL JS map rendering.
- Integrated OpenWeatherMap APIs to sample weather conditions along route waypoints in real time.
- Implemented Netlify proxy functions for secure API key handling, with planned integration of LLM API services for route weather safety advice.

Multi-Modal Clinical Phenotyping System PyTorch FastAPI FHIR R4 Healthcare AI Project

- Built a dual-encoder PyTorch pipeline pairing BioClinicalBERT for unstructured clinical notes with a transformer for lab test features on MIMIC-IV patient records.
- Developed a FastAPI inference service with SHAP lab feature importance, BERT attention token scores, FHIR R4 bundle adapters, and Gradio web interface.

PROFESSIONAL EXPERIENCE

Software Developer
GE Healthcare (Bengaluru, India)

- Developed internal license management software for hospital equipment (MRI and CT scanners), enabling staff to activate and update feature licenses.
- Built automated installation and upgrade tooling to reduce manual overhead for active hospital scanners.
- Collaborated with engineering teams to deliver annual software feature updates and security patches.

Software Developer

Swiss Re (Hyderabad, India / Zurich, Switzerland)

- Developed *SwiftMX*, a transaction conversion tool mapping business team requests into automated bond and security purchase orders via third-party financial tools.
- Built testing and data migration utilities for TRAX application upgrades to verify configuration transfer success rates.
- Implemented custom encryption middleware routing payment files (salary and insurance disbursements) securely to banking institutions.

Junior Software Engineer

American Express (Hyderabad, India)

- Developed online PDF credit statement generation features for American Express Mexico internet banking.
- Participated in software development lifecycle: requirements review, component development, testing, and production deployment.

CERTIFICATIONS & LANGUAGES

Certifications: Python Institute Certified Programmer | Lean Six Sigma Yellow Belt | TOEFL iBT (Score: 98/120)

Languages: English (Fluent / Professional), Telugu (Native), Danish (Prøve i Dansk 2 / PD2 bestået: B1 reading and writing, actively developing speaking and listening)